

AI MODEL BIAS & FAIRNESS AUDIT

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Model: Random Forest Classifier | Dataset: Adult Income (UCI)

1. Model Performance Summary

Accuracy: 0.8609

Precision: 0.7458

Recall: 0.6423

2. Bias Audit Findings

Finding 1: Sensitive Attribute - SEX

HIGH

Demographic Parity Difference (DPD): 0.1841

Equalized Odds Difference (EOD): 0.1091

Interpretation: Meaningful disparity across sex. Outcomes are not equitable - recommend retraining with fairness constraints applied.

Finding 2: Sensitive Attribute - RACE

CRITICAL

Demographic Parity Difference (DPD): 0.2

Equalized Odds Difference (EOD): 0.1097

Interpretation: Meaningful disparity across race. Outcomes are not equitable - recommend retraining with fairness constraints applied.

3. Model Explainability (SHAP)

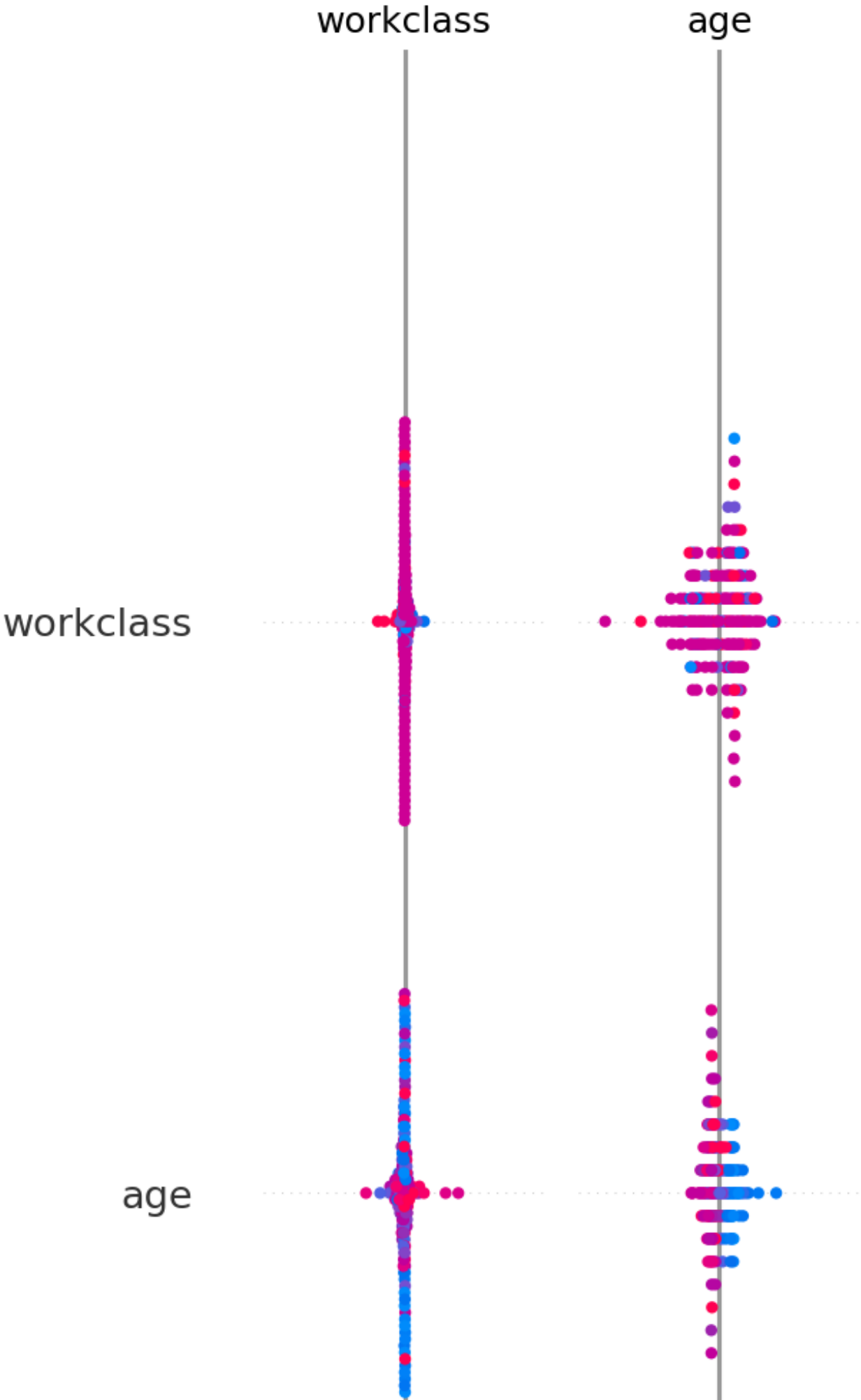


Figure 1: SHAP feature importance - features pushing model decisions toward high/low income prediction.

4. Recommendations & Remediation

1. Apply fairness constraints during model retraining (e.g. ExponentiatedGradient).
2. Remove or proxy-blind sensitive attributes from feature set where legally required.
3. Implement ongoing fairness monitoring - re-audit model quarterly.
4. Escalate CRITICAL/HIGH findings to model risk committee for sign-off.
5. Document audit trail and findings in the model risk register.